

Survey Wave Explorer

PURPOSE

- The main goal of this Shiny App is to showcase response trends across waves, utilizing statistical methods such as Ordinal Categorical Analysis and Vectorized General Linear Models.
- The secondary goal is for survey analysts to showcase to an audience how responses change across waves, using intuitive knowledge of country dynamics or policy changes to draw effective conclusions.

FEATURES

DATA UPLOAD

The Data Upload section is the “homepage” of the Shiny App and the first user interaction. In this section users upload a .sav file with a numeric unlabeled Wave column.

- For example, for a correct file the column name is Wave and values are; 1, 2,3 etc.
- For a wrong file the column name is wave but values are; Wave 1, 2022, Wave 2 2025 etc.
- This section gives a preview of the dataset and a numerical summary or frequency table based on the variable selected
 - Categorical/ Binary Variables: Frequency table counting variables per level as well as NA's
 - Variables with less than or equal to 10 unique values are coded as categorical.
 - Numerical Variables: Not many numerical variables in test dataset so this gives a numerical summary with the unweighted Median, Mean, as well as the minimum and maximum numbers in that variable

- “Select Wave” allows users to filter the dataset preview by Wave. This preview is available for download as well.

WAVE COMPARISON

GRAPH SECTION

This tab allows users to select waves to see the proportion of responses for each level.

- The graph can be an inclusion of the proportion of responses for all waves, or the user can choose to simplify by deselecting waves and looking at different intervals.
- “Select Variable” allows users to choose which variable they want to view, and the legend on the left side of the graph reflects the response choices the respondents had to choose from.
- “Select Graph type” Allows user to choose the type of graph downloaded, the options are Stacked Bar Graph, Line Graph and Grouped Bar Graph.
 - o Based off of research, these are the most efficient in showcasing per-level-counts of categorical variables
- The Wave comparison tab can also be used to track numeric variables. By choosing either a line graph or ‘grouped graph’ it demonstrates how the mean response for a numeric variable fluctuates across waves.
 - o Choosing a stacked bar graph will result in an error message and a hint to choose one of the other two working graphs.

MISSINGNESS SUMMARY TABLE

This section of the tab is a summary data table of the percentage of responses and Na’s that are coded as missingness.

- The goal for this tab was to give a summary of what is seen in the graph. When analyzing a variable longitudinally across waves its important for users to see if there’s an effective sample size of responses for that variable across waves.
- The partitions of the table are %missing, True NA, # Refused, #Dont Know and the total respondents for that wave.
 - o If any Wave has over 5% missing, I would make the executive decision to deselect that Wave from being compared in the analysis.

NON- RESPONSE ANALYSIS

- The non-response analysis section in the Waves Comparison tab was added following feedback. It would be pivotal for analysts to track if the proportion of nonresponse (i.e. “Don’t Know”, “Refused”) changes across waves significantly.
- Since this is unnecessary in the context of finding significant differences between waves, the user can choose to view it alongside the priorly rendered graph or not.
- This function in the Shiny App uses the `prop.test` function in base R to test the null hypothesis that the proportions of non-response are the same across waves.
- Selecting and deselecting waves in the Wave Comparison tab changes the statistics shown when hitting this button.
 - o The test provides a Chi-square test statistic, the degrees of freedom (1 - # of Waves selected), and a p-value.
 - A p-value of less than 0.05 shows that there is a significant difference in the proportion of non-response between waves.

ORDINAL CATEGORICAL ANALYSIS

Ordinal Categorical Analysis is a type of statistical analysis that compares ordinal variables (variables with a hierarchy i.e. Likert scales) against a certain predictor.

- In this statistical model a categorical survey question is the ordinal variable and Wave is the predictor.
- The model inevitably shows if there is a statistically significant relationship between Survey question.
- Statistically significant Wave Variable pairs have their p-values highlighted in gray
 - o Where do conclusions come into? Well, the odds ratio shows the extent of the relationship. Since a ordinal (hierarchal) dynamic between levels is assumed for the model, The odds ratio tells us the direction of the relationship.
 - An odds Ratio less than 1 means: The odds of being in a higher category are OR times the baseline wave, the odds decrease by about $(1 - OR)\%$. Show example
 - An odds ratio greater than 1: The odds of being in a higher category are OR times the baseline wave, the odds of choosing in a higher ordinal category increase by $(1 - OR) \%$

ASSUMPTIONS

- Now as with all statistical tests there are assumptions the data must follow
 - o Most of the assumptions are satisfied simply off the way D3 collects survey data, the only one of worry is the Proportional Odds Assumption, which the shiny is trained to identify if its satisfied or not.
 - In the case of satisfaction the user can move onto the interpretations section of the app
 - In the case of dissatisfaction the user can move onto fitting the glm model in the next tab.
- Waves can also be selected or deselected, the lowest wave number selected will always be the baseline your comparing everything else to. Give Example

BATCH ANALYSIS

GENERALIZED CATEGORICAL ANALYSIS (GLM)

In the case that the assumptions check on OCA says to move on the same model is fit but now showcasing inter-response differences.

- Significant pairs are still highlighted in grey and high and low odds ratios according to the statistical threshold are highlighted in orange and blue respectively.
- The key difference for this tab is interpretation. Since the proportional hazards assumption is relaxed, now the odds for each level are shown so one doesn't have to assume what the higher ordinal category is versus the lower one.
- Odds Ratios for this test is interpreted as the odds of choosing a level greater than the next level up is (OR) times the baseline. Show example: wave 28 the odds of choosing a level ≥ 2 was 6.85 times higher than the baseline, so basically in terms of the survey there was a sharp decline in people choosing "very good" to rate their supply of electricity.

ASSUMPTIONS

- As this is a diff statistical test it has different assumptions and all are satisfied except the STD error assumption.

- The shiny automatically flags and inflated standard errors and shows at the bottom if interpretations can be drawn from the data or not.
- Just like the last tab waves can also be selected/deselected. To improve access, it is pretty pivotal that if a question isn't asked in a wave to remove it because it can influence things and lead to wrong interpretations.
- Also interpretations may be hard to make if there's limited statistical knowledge so that's what the next tab helps with.

BATCH ANALYSIS

INTERPRETATIONS

PROBABILITIES PLOT & TABLE

- In this section its important that you choose the test type with satisfied assumptions so if its OA choose that and if its GLM choose that.
- This tab shows a predictive plot which plots the probability of a respondent choosing a response for each wave.
 - This is important because users and clients can use an accurate model in order to see how responses to a certain var fluctuate across waves. They can also input their own knowledge to assume causation, for example maybe there was a policy change for electricity so in wave 18 people feel good about their power versus how people felt about power before the policy changed in wave 17.
 - It's also hard to maybe visualize probability based off the graph so the probability table at the bottom is the remedy for that.
 - For example, the probability of a respondent choosing very good for CndPower is 33.3% in Wave 19 but in wave 20 that probability drops to 14.7%.
- The text interpretation helps users with interpreting complex Odds ratios by automating the statistical interpretation for the user. The highest and lowest odds ratios of the selected model are interpreted.
- This tab is limited in the fact that the variable of interest must be defined in the OA or GLM tab first.

TEXT INTERPRETATIONS

- There are two sets of automated text interpretations, one set coded specifically for the ordinal regression model, and the other coded specifically for the vectorized GLM.
 - o The point of making them separate is that each model shows two separate things, though alluding to the same general principle of analyzing trends across waves.
- The ordinal interpretations basically demonstrate a shift in the response odds across the different Wave levels and highlights the most likely outcome of each wave, so one could note it before analyzing the probabilities table
- The vglm interpretations show the likelihood or odds of moving to a higher ordinal category, still showing between-wave differences but gives a bit more of a specific interpretation.

GENERATE REPORT

CHOICE REASONING

UI CHOICES

- Making all the graphs using the D3 company color pallet
- Tabs, buttons and fonts are also consistent with D3 styling choices
- After working over the summer, I noticed a high number of categorically coded variables, which is why I focused on consistently analyzing those across waves instead of analyzing binary and numeric variables as well.
- A ton of user input on each tab allows more flexibility in what a User wants to see while clicking through tabs.
- Bookmarking: While clicking through other tabs, the other tabs stay on the selected inputs for efficiency.
- I chose a line graph because I felt that's the most visually appealing for probabilities.

STATISTICAL CHOICES

ORDINAL CATEGORICAL REGRESSION ANALYSIS

Ordinal Categorical Regression is the same concept as logistic regression only involves the assumption of ordinality amongst variables.

- The function I decided to use to create the model was the `clm()` function from the 'ordinal' package which is completely dedicated to analyzing ordinal data such as Likert Scales. It also has a very flexible function, doing well at handling random effects (since data is cross-sectional instead of panelist), whereas other ordinal regression functions such as `polr()` focus mainly on proportional odds model.
- The main goal of the dashboard is to show how responses change across wave, doing ordinal categorical analysis hones this question a little more and instead asks, "How likely is someone to choose a higher response level", which is a little bit better and more complex.
 - o The function creates a simple categorical regression model, so there's no need to test multicollinearity. Using the chosen survey question as the response and wave as a predictor, the model is fit.
 - o This test is perfect when the dependent variable has ordered categories such as low, medium, and high.
 - o All assumptions are satisfied except for the proportional odds assumption (this assumption assumes that the effect of each predictor is the same across all thresholds.)
 - Brant test is how this assumption is checked, the Shiny automatically does the brant test and prints out a statement at the bottom of whether its satisfied. I believe this was right to include because in the case of a user making interpretations about a test output, someone can go in and double check whether it's accurate instead of having to test it themselves.
 - One thing I recently noticed is that sometimes when you look at smaller wave intervals such as Waves 28-30 vrs Waves 19-30, OCA handles it pretty well and assumptions are met.
- I chose to also include z-value and std-error even though the data is unweighted, because the examination here is between Waves, not between respondents, so it can be deemed accurate in the context of these interpretations.

VECTOR GENERALIZED LINEAR MODELING

In the case that the effect of predictor (Wave) changes across variables, a Vector Generalized Linear Model is fit which relaxes the proportional odds assumption, making it a model constraint rather than a structural limitation on the model being fit.

- This works by fitting a non-proportional odds or cumulative model where each level of the variable gets its own coefficient.

- Instead of fitting all levels of the predictor at once against all ordinal outcomes as in `clm()`, `vglm` fits each predictor:response pair one at a time against the reference baseline category
 - o In `vglm` usually the reference level is the “last level”, so in the table that’s why it seems like it might be missing a threshold.
- This model is also very efficient in determining the extent of differences across categories of the ordinal variable, rather than just determining whether there’s a significant difference between waves.
 - o The tradeoff for this model is sometimes a loss of power depending on sample size, especially since estimating extra points means wider confidence intervals. (I personally didn’t see this as a huge tradeoff, but others more familiar with this concept in practice might).
- A standard MLM might have also been the remedy for this, but it doesn’t control the random effects seen in the data.
- The assumptions worth noting in the app for this model is the case that the chosen variable exhibits Quasi-complete separation which is also known as a predictor ‘perfectly’ predicting a response variable.
 - o This could be checked automatically (Which in the Shiny it is) by checking for an inflation of Std. Errors. The constraint I specifically added was that if a std error for a wave for a particular variable is over 3, then the assumption is flagged as having been violated.

PREDICTED PROBABILITY PLOT & TABLE

- Including a predicted probabilities plot is efficient in showcasing how the predicted likelihood of an event (in this case the event is choosing a particular level of the chosen categorical variable) changes across survey periods.
- The table quantifies the predicted probabilities seen in the plot, so one doesn’t have to play a guessing game or overly zoom in.
- I chose to create a line graph for this specifically because it is more visually appealing and easy to interpret given the context.
 - o For example, flat lines mean barely any change across time; curvy lines mean the opposite.
 - o The first wave in the graph is the base probabilities or baseline likelihood for each group, with every weight these changes. The vertical distance also shows the predicted effect size. Furthermore, if the lines cross at any point, it shows an interaction effect between the levels of the variable.

OVERALL

Overall, the choices I made in this Shiny App are meant to help analysts be able to track trends in response to their questions across waves. This helps not only with delivering extra knowledge to the client but also aids in knowing whether a question is still a good fit for a survey. For example, if the proportion of those that don't answer continues to increase across each wave it might be worth noting that it's not working very well in field. Another example mentioned earlier is using the knowledge of the country to deliver actionable insights alongside the plot. For example, if they changed a policy that impacted a lot of people in that Nation, it may be worth noting how the likelihood of choosing a certain response option changed before and after the implementation.

FUTURE IMPROVEMENTS

- Allow Users to choose the type of probability plot, with or without std errors
- See if weights influence statistics since everything is unweighted
Add tests that analyze/interpret numeric and binary variables
- More validation/pop-up messages throughout the case of errors.

TEAM FEEDBACK

- Figure out how to add weights per wave (Some merge files have a weight column, and I feel like this could be utilized some way.)
- Have AI better interpret analysis tests and give most significant effects in understandable language.
- Figure how to handle large amounts of missingness ~ VGLM already does this pretty well.
- Find more analysis' to test numeric variables across waves
 - o For binary variables try and input a Bayesian beta binary test into app
 - o Showcase type of variable upon selection

SOURCES

- [Shiny for R Gallery](#)
- <https://stats.oarc.ucla.edu/other/mult-pkg/faq/general/faqwhat-is-complete-or-quasi-complete-separation-in-logistic-regression-and-what-are-some-strategies-to-deal-with-the-issue/>

- <https://stats.stackexchange.com/questions/233687/why-do-we-plot-observed-probability-vs-predicted-probability-when-calibrating-a>
- https://cran.r-project.org/web/packages/jtools/vignettes/effect_plot.html
- <https://metricgate.com/docs/brant-test-proportional-odds/>
- <https://metricgate.com/blogs/common-mistakes-in-ordinal-logistic-regression/>
- <https://druedin.com/notes/predicted-probabilities-in-r.html>
- <https://hbiostat.org/rmsc/ordinal> <- math behind things
- <https://docs.posit.co/connect-cloud/how-to/r/llm-shiny-r.html>
-